

CKM Matrix from Left-Right Symmetric Weak-Isospin Geometry and Chiral Amplitude

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Abstract

We present a geometric construction of the Cabibbo–Kobayashi–Maskawa (CKM) quark-mixing matrix magnitudes based on the left-right (LR) symmetric weak-isospin geometry. The construction is built on two internal structures: (i) the LR isospin circle

$$(T_3^L)^2 + (T_3^R)^2 = \frac{1}{4}$$

in the (T_3^L, T_3^R) weak-isospin plane, and (ii) the weak-isospin activity operator

$$\hat{\Gamma} = \mathbb{I} - \hat{T}_3^L,$$

which yields exact, parameter-free amplitude factors for five of the six quark flavours. The right-handed weak-isospin coordinate T_3^R , which arises naturally in left-right symmetric extensions of the Standard Model and is geometrically orthogonal to T_3^L in the (T_3^L, T_3^R) plane, provides internal coordinates for right-handed quark states and establishes the chiral hierarchy of CKM transition amplitudes. A single constitutive deformation parameter $\delta_b = 0.0665$ describes the departure of the bottom-quark T_3^R value from the LR-symmetric configuration. This deformation introduces anisotropic LR isospin projections with semi-axes

$$a = \frac{1}{2} - 2\delta_b, \quad b = \frac{1}{2} + \delta_b,$$

satisfying the linear constitutive constraint $a + 2b = \frac{3}{2}$. Two inputs extracted from PDG 2024 adjacent-generation ratios [8], $(\lambda_{12}, \lambda_{23})$, together with the constitutive saturation relation (derived in the Appendix)

$$\delta_b(1 - \delta_b) = \frac{7}{6}\lambda_{12}^2$$

are sufficient to generate all nine CKM magnitudes through the relation

$$|V_{ij}| = \frac{\sqrt{\Gamma_i \Gamma_j} S_{ij} C_{ij}}{\mathcal{N}_i}.$$

The resulting CKM magnitudes agree with PDG 2024 values within $\mathbf{0.51\sigma}$. The two non-input predictions $|V_{ub}|$ and $|V_{td}|$ agree with experiment at 0.51σ and 0.21σ , respectively. Two derivations appended to the main construction extend these results: Appendix A derives δ_b analytically from a constitutive saturation law rooted in the LR charge geometry, reducing the independent inputs from three to two; Appendix B establishes that the Wolfenstein Cabibbo parameter $\lambda = 1/(\pi\sqrt{2}) \approx 0.22508$ follows from the azimuthal projection of the LR isospin circle, matching the PDG value to 0.03% without any CKM input, and pointing toward a single-input construction. The proposed construction indicates that the observed hierarchy of quark flavour mixing can emerge from the geometry of left-right symmetric weak-isospin coordinates, chiral amplitude structure, and generation-transition suppression, rather than being encoded solely in independent phenomenological parameters.

1 Introduction

The Cabibbo–Kobayashi–Maskawa (CKM) matrix [3, 4] provides the amplitudes of charged-current flavour transitions between up-type and down-type quarks. The transitions from up-type quarks (u, c, t) to down-type quarks (d, s, b) are represented by

$$V_{\text{CKM}} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}.$$

The conjugate transpose $V_{\text{CKM}}^\dagger$ describes the corresponding reverse charged-current transitions. For three quark generations, the CKM matrix contains four independent physical parameters: three mixing angles and one CP-violating phase. These parameters are determined by global fits to experimental data [8]. Although the CKM matrix provides a precise mathematical description of quark flavour mixing, the origin of the observed hierarchy and the numerical values of its parameters remain unexplained within the Standard Model.

The present work investigates this question through an internal quantum-state construction based on the left-right (LR) symmetric weak-isospin decomposition. In LR symmetric extensions of the Standard Model [6, 7], right-handed quarks form doublets under $\text{SU}(2)_R$ and carry a definite right-handed weak-isospin projection $T_3^R = \pm\frac{1}{2}$. The present work exploits this T_3^R quantum number as a geometric coordinate for right-handed quark states, orthogonal to the Standard-Model weak-isospin projection T_3^L . In the Standard Model, right-handed quarks are $\text{SU}(2)_L$ singlets with $T_3^L = 0$ and possess no natural internal coordinate. The T_3^R axis resolves this by assigning right-handed states to the imaginary axis of the complex LR coordinate $T_3 \equiv T_3^L + iT_3^R$.

The two weak-isospin projections, T_3^L and T_3^R , define a two-dimensional quantum space in which quark states occupy specific locations on an LR isospin circle of radius $\frac{1}{2}$. The resulting structure provides a unified description of left-handed and right-handed quark states within the same internal space. The central contribution of this work is the identification of the weak-isospin activity operator

$$\hat{\Gamma} = \mathbb{I} - \hat{T}_3^L$$

as the quantity controlling the relative coupling amplitudes responsible for CKM mixing. The operator yields parameter-free amplitude factors for the up-type and down-type quark sectors and establishes the amplitude hierarchy from the internal quantum-state geometry. The remaining flavour-suppression factors are obtained from adjacent-generation transition ratios and combined through a compositional construction to generate the complete CKM magnitude matrix.

The anisotropic deformation of the internal state space provides the mechanism for incorporating the third-generation mass effect and generating the generation-skip transition factors. The resulting formulation produces all nine CKM magnitudes from three physically motivated inputs and allows a direct comparison with experimental values.

The paper is organised as follows. Section 2 reviews the standard CKM framework. Section 3 introduces the LR isospin geometry and quark state assignments. Section 4 develops the weak-isospin activity operator and derives the amplitude factors. Section 5 extracts the mixing inputs from PDG 2024 ratios [8]. Section 6 presents the anisotropic LR isospin metric and generation-skip projection factors. Section 7 presents the master formula and CKM construction. Section 8 compares the results with PDG 2024 values. Section 9 discusses the geometric interpretation, CP-violation sector, and future parameter reduction. Section 10 concludes the paper. Appendix A provides the parameter accounting and reduction roadmap.

2 The CKM Matrix: Standard Framework and its Gap

In the Standard Model, quark masses originate from Yukawa couplings to the Higgs field [8]. Following spontaneous symmetry breaking, the complex up-type and down-type quark mass matrices are diagonalised by independent unitary rotations V_L^u and V_L^d acting on the left-handed doublets. The CKM matrix is the mismatch between these two rotations:

$$V_{\text{CKM}} = V_L^u (V_L^d)^\dagger, \quad (1)$$

which governs the charged-current weak interaction:

$$\mathcal{L}_{\text{CC}} = -\frac{g}{\sqrt{2}} \bar{u}_{L,i} \gamma^\mu (V_{\text{CKM}})_{ij} d_{L,j} W_\mu^+ + \text{h.c.} \quad (2)$$

Because V_{CKM} is unitary, rephasing of quark fields reduces its 18 real parameters to 4 physical observables: three mixing angles and one CP-violating phase [8].

The Wolfenstein parametrisation [5] makes the observed generation hierarchy explicit:

$$V_{\text{CKM}} = \begin{pmatrix} 1 - \frac{1}{2}\lambda^2 & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \frac{1}{2}\lambda^2 & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{pmatrix} + \mathcal{O}(\lambda^4), \quad (3)$$

with PDG 2024 values $\lambda = 0.22501$, $A = 0.826$, $\rho = 0.159$, $\eta = 0.348$ [8]. The hierarchy $|V_{ij}| \sim \lambda^{|i-j|}$ is clearly established experimentally. However, the Standard Model treats λ , A , ρ , and η as consequences of the unknown Yukawa coupling matrices and provides no derivation of their numerical values from an underlying principle.

The present work investigates whether the CKM hierarchy can emerge from an internal LR weak-isospin geometry. The projections T_3^L and T_3^R define an internal space whose geometric structure determines the amplitude factors and transition suppression, producing the CKM magnitude matrix from a small number of physically motivated inputs.

3 Weak-Isospin Geometry and (T_3^L, T_3^R) Isospin Plane

In the Standard Model, weak interactions are governed by the $\text{SU}(2)_L$ gauge symmetry, which acts only on left-handed fermions. Left-handed quarks and leptons transform as $\text{SU}(2)_L$ doublets with weak-isospin projections $T_3^L = \pm \frac{1}{2}$ for the up-type and down-type members of each doublet, whereas right-handed fermions are $\text{SU}(2)_L$ singlets and therefore satisfy $T_3^L = 0$. Since no right-handed weak-isospin symmetry exists, $T_3^R = 0$ for all Standard Model fermions. Consequently,

$$(T_3^L)^2 + (T_3^R)^2 = \begin{cases} \frac{1}{4}, & \text{left-handed states,} \\ 0, & \text{right-handed states.} \end{cases}$$

3.1 Left-Right Symmetric Framework

The Standard Model treats the two chiralities asymmetrically by assigning weak interactions exclusively to left-handed fermions, while the Left-Right (LR) symmetric theories restore this symmetry at high energies by introducing an additional $\text{SU}(2)_R$ gauge symmetry under which right-handed fermions also transform as weak-isospin doublets. The electroweak gauge group is thereby extended to

$$\mathcal{G}_{\text{LR}} = \text{SU}(3)_C \otimes \text{SU}(2)_L \otimes \text{SU}(2)_R \otimes U(1)_{B-L}, \quad (4)$$

where left-handed fermions transform as $(2, 1)$ and right-handed fermions as $(1, 2)$ under $SU(2)_L \times SU(2)_R$. Within this framework, the electric charge operator assumes the familiar form

$$Q = T_3^L + T_3^R + \frac{B - L}{2}, \quad (5)$$

where T_3^L and T_3^R denote the third components of the left- and right-handed weak isospin, respectively.

In the present work we adopt the standard left-right quantum-number assignment solely as the kinematic foundation for a geometric construction in the (T_3^L, T_3^R) plane. No explicit assumptions are made regarding the dynamics of the LR gauge theory, the Higgs sector, or the mechanism of spontaneous symmetry breaking. Instead, only the quantum-number decomposition embodied in Eq. (5) is employed, allowing (T_3^L, T_3^R) to serve as the fundamental coordinates for the hyperspin geometry developed in the following sections.

3.2 Complex Representation of the LR Weak-Isospin State

Left-right symmetric theories contain two independent weak-isospin sectors associated with $SU(2)_L$ and $SU(2)_R$. The third components T_3^L and T_3^R therefore constitute two independent real coordinates describing the complete left-right weak-isospin state. From a geometric viewpoint, it is natural to represent this ordered pair by a single complex coordinate,

$$T_3 \equiv T_3^L + iT_3^R, \quad (6)$$

where the real part $\text{Re}(T_3) = T_3^L$ and imaginary part $\text{Im}(T_3) = T_3^R$ correspond to the left- and right-handed weak-isospin projections, respectively. This definition is introduced as a compact geometric representation of the combined LR weak-isospin state; it does not modify the underlying gauge-theoretic structure.

The complex representation provides a natural mathematical framework for the CKM construction. Since the CKM matrix is a unitary complex matrix, representing the LR weak-isospin coordinates on a complex plane directly facilitates the subsequent geometric construction of the Wolfenstein parameters and the CKM mixing amplitudes.

It is important to emphasise that the complex representation $T_3 = T_3^L + iT_3^R$ does not by itself introduce or imply CP violation. Rather, it provides a natural two-dimensional geometric space in which flavour mixing can be represented. The experimentally observed CP-violating phase appears subsequently as a phase associated with the mapping from the normalised LR weak-isospin circle to the Wolfenstein parametrization of the CKM matrix, through the identification $\bar{\rho} + i\bar{\eta} = s T_3$ (Section 6).

Within the present construction, individual left- or right-handed quark states carry one of the assignments $T_3^L = \pm \frac{1}{2}$ (left-handed) or $T_3^R = \pm \frac{1}{2}$ (right-handed), with the complementary component zero. The symmetric LR isospin geometry is therefore characterised by

$$(T_3^L)^2 + (T_3^R)^2 = \frac{1}{4}, \quad (7)$$

which defines the *LR isospin circle* of radius $\frac{1}{2}$ in the (T_3^L, T_3^R) plane. Equivalently, $|T_3| = \frac{1}{2}$. This relation characterises the undeformed reference configuration of the internal state space; deviations are associated with mass-dependent effects in the third generation.

3.3 Quark State Assignments and Chirality Rule

The two axes serve complementary roles determined by the chiral structure of the weak interaction.

Chirality rule. Left-handed fermions carry $SU(2)_L$ weak isospin and occupy the real axis: $(T_3^L, T_3^R) = (\pm \frac{1}{2}, 0)$, $T_3 \in \mathbb{R}$. Right-handed fermions are $SU(2)_L$ singlets but carry $SU(2)_R$ weak

isospin and are assigned to the imaginary axis: $(T_3^L, T_3^R) = (0, \pm \frac{1}{2})$, $T_3 \in i\mathbb{R}$. The T_3^R axis provides an internal geometric coordinate for right-handed quark states, which in the Standard Model have $T_3^L = 0$ and no corresponding weak-isospin coordinate.

3.4 Quark State Assignments and the b -Quark Deformation

Applying the chirality rule together with the $SU(2)_L$ assignments $T_3^L(u_L) = +\frac{1}{2}$ and $T_3^L(d_L) = -\frac{1}{2}$, the right-handed quark states in the symmetric chiral limit are assigned the following LR isospin coordinates:

$$u_R, c_R, t_R: \quad T_3^R = +\frac{1}{2}, \quad T_3 = +\frac{i}{2}, \quad (8)$$

$$d_R, s_R: \quad T_3^R = -\frac{1}{2}, \quad T_3 = -\frac{i}{2}. \quad (9)$$

The d and s quarks share an identical T_3^R coordinate in this symmetric configuration. This geometric degeneracy provides the basis for Cabibbo universality, which is verified quantitatively in Section 5.1.

The b quark departs from the symmetric chiral configuration. With pole mass $m_b \approx 4.18 \text{ GeV}$ [8], the third-generation down-type state experiences a mass-dependent deformation of its internal coordinate. The effective right-handed isospin coordinate is displaced by $\delta_b > 0$:

$$b_R: \quad T_3^R = -\left(\frac{1}{2} + \delta_b\right), \quad T_3 = -i\left(\frac{1}{2} + \delta_b\right), \quad \delta_b = 0.0665. \quad (10)$$

The u , d , s , c , and t states define the symmetric reference configuration on the LR isospin circle, while the b state introduces the first asymmetric deformation. The quark state assignments are summarised in Table 1.

Table 1: Quark state assignments in the (T_3^L, T_3^R) LR isospin plane. Left-handed states lie on the T_3^L axis; right-handed states on the T_3^R axis in the symmetric configuration. The b_R state represents an asymmetric deformation with $\delta_b = 0.0665$.

Row	State(s)	T_3^L	T_3^R	$T_3 = T_3^L + iT_3^R$	Assignment
1	u_L, c_L, t_L	$+\frac{1}{2}$	0	$+\frac{1}{2}$	T_3^L axis (left)
2	d_L, s_L, b_L	$-\frac{1}{2}$	0	$-\frac{1}{2}$	T_3^L axis (left)
3	u_R, c_R, t_R	0	$+\frac{1}{2}$	$+\frac{i}{2}$	T_3^R axis (right)
4	d_R, s_R	0	$-\frac{1}{2}$	$-\frac{i}{2}$	T_3^R axis (right)
5	b_R	0	$-\frac{1}{2} - \delta_b$	$-i(\frac{1}{2} + \delta_b)$	T_3^R axis (deformed)

4 Weak Isospin Activity and Background Stabilization

4.1 Gell-Mann–Nishijima Decomposition and the Two Constitutive Operators

Both constitutive operators of the present construction emerge from a single fundamental relation. The Gell-Mann–Nishijima (GNN) formula [1, 2]

$$Q = T_3 + \frac{Y}{2} \quad (11)$$

separates the electric charge into a *dynamical* component T_3 (coupling to the charged weak current) and a *background* component $Y/2$ (the electroweak hypercharge, independent of the

transition). These two components motivate two constitutive operators — the central pillars of the present construction — one for each part of the charge.

The Weak Isospin Activity Operator $\hat{\Gamma}$ is defined as the complement of the dynamical T_3 component:

$$\hat{\Gamma} \equiv \mathbb{I} - \hat{T}_3^L. \quad (12)$$

It quantifies the *weak isospin activity* of a quark state: the portion of the unit weak isospin that is not locked into the current isospin projection and is therefore available to drive flavor-changing transitions through the charged weak current.

The Background Stabilization Factor K is defined as the complement of the background $Y/2$ component:

$$K \equiv 1 + \frac{Y}{2}. \quad (13)$$

It quantifies the structural capacity of the electroweak background to amplify and sustain the geometric deformation of the isospin configuration that results from flavor mixing. Its derivation from the LR charge geometry is given in Appendix C.

The two operators are not independent postulates. Their sum obeys an algebraic identity that follows directly from Eq. (11); since T_3 cancels exactly,

$$Q + \Gamma = \left(T_3 + \frac{Y}{2}\right) + (1 - T_3) = 1 + \frac{Y}{2} = K. \quad (14)$$

For all six quark flavours in the symmetric (undeformed) sector, $K = Q + \Gamma = 7/6$ universally (Table 2), establishing K as a *quark electroweak invariant* fixed entirely by the GNN charge structure — not a free parameter.

Table 2: Universality of $K = Q + \Gamma$ across all quark flavours in the symmetric (undeformed) sector. The cancellation of T_3^L in Eq. (14) ensures $K = 7/6$ independently of quark flavour or generation.

Quark(s)	T_3^L	Q	$\Gamma = 1 - T_3^L$	$Q + \Gamma$
u, c, t	$+\frac{1}{2}$	$+\frac{2}{3}$	$\frac{1}{2}$	7/6
d, s, b	$-\frac{1}{2}$	$-\frac{1}{3}$	$\frac{3}{2}$	7/6

4.2 The Weak Isospin Activity Operator

The eigenvalue of $\hat{\Gamma}$ for quark flavour q is the *weak isospin activity factor* $\Gamma_q = 1 - T_3^L(q)$. Its physical content follows directly from the LR isospin assignments.

For *up-type quarks* ($T_3^L = +\frac{1}{2}$), the weak-isospin projection is at its positive maximum, leaving only half the unit isospin uncommitted to the projection:

$$\Gamma_{u,c,t} = 1 - \frac{1}{2} = \frac{1}{2}. \quad (15)$$

These states are *half-active*: one half of the unit weak isospin is already engaged in the isospin projection, and only the remaining half is free to participate in flavor-changing transitions.

For *down-type quarks* ($T_3^L = -\frac{1}{2}$), the projection is at its negative maximum, so the complement exceeds unity:

$$\Gamma_{d,s} = 1 - \left(-\frac{1}{2}\right) = \frac{3}{2}. \quad (16)$$

These states are *over-active*: the negative isospin projection adds to rather than depletes the transition capacity, producing an activity factor 50% larger than the unit value.

The ratio $\Gamma_{d,s}/\Gamma_{u,c,t} = 3$ establishes a *three-to-one* activity hierarchy between down-type and up-type quarks. This asymmetry is the primary geometric origin of the CKM hierarchy: the geometric mean $\sqrt{\Gamma_i \Gamma_j}$ entering the master formula gives

$$\sqrt{\Gamma_u \Gamma_d} = \sqrt{\frac{1}{2} \cdot \frac{3}{2}} = \frac{\sqrt{3}}{2} \approx 0.866 \quad (17)$$

for the dominant 1–2 and 2–3 transitions, providing the leading amplitude before any generation-mixing suppression is applied.

A defining constitutive correspondence of the LR construction is that left-handed and right-handed weak-isospin projections are equal in the symmetric sector: $T_3^L(q) = T_3^R(q)$. This allows the activity factor to be expressed equivalently through the right-handed coordinate:

$$\Gamma_q = 1 - T_3^L(q) = 1 - T_3^R(q). \quad (18)$$

This dual representation makes the role of the T_3^R axis explicit: the right-handed isospin position encodes the transition activity of the associated left-handed state, despite the absence of right-handed charged currents in the Standard Model low-energy limit.

For the b quark, the LR correspondence is deformed: $T_3^R(b_R) = -(\frac{1}{2} + \delta_b)$, so $\Gamma_b = \frac{3}{2} + \delta_b$. The b -quark deformation δ_b is the sole source of inter-generational asymmetry in the construction, and its saturation by the Born transition probability λ_{12}^2 (Appendix A) connects the activity operator directly to the Cabibbo mixing scale hierarchy.

4.3 The Background Stabilization Factor

The Background Stabilization Factor $K = 1 + Y/2 = 7/6$ plays a role complementary to $\hat{\Gamma}$: where $\hat{\Gamma}$ governs the amplitude of a flavor-changing transition, K governs the structural response of the isospin geometry to a transition that has occurred.

K is a quark electroweak invariant. As established by Eq. (14) and Table 2, $K = Q + \Gamma = 7/6$ for all six quark flavours: the isospin projection T_3 cancels algebraically, leaving a value determined solely by $Y/2$. For quarks, $(B - L)/2 = 1/6$ (with $B = \frac{1}{3}$, $L = 0$), so the increment $K - 1 = \frac{1}{6}$ is precisely the baryon-number hypercharge contribution in Eq. (5), connecting K directly to the $U(1)_{B-L}$ sector of the LR gauge group.

K amplifies the Born transition probability into a structural deformation. By the Born rule, the probability for the leading Cabibbo flavor transition is $P_{12} = |\lambda_{12}|^2 = \lambda_{12}^2$. The Background Stabilization Factor converts this probability into the geometric deformation of the b -quark right-handed isospin coordinate through the constitutive saturation law (Appendix A):

$$\delta_b(1 - \delta_b) = K \lambda_{12}^2 = \frac{7}{6} \lambda_{12}^2. \quad (19)$$

Because $K > 1$, the electroweak background amplifies the structural deformation beyond the naive Born probability. The amplification factor $K - 1 = \frac{1}{6} = (B - L)/2$ reveals that this excess originates in the $U(1)_{B-L}$ sector.

K does not generate transitions. Unlike $\hat{\Gamma}$, which directly determines the coupling amplitude of a charged-current transition, K modulates the *structural geometry* of the LR isospin configuration without itself driving any flavor change. The full derivation of K from the T_3^R axis charge geometry, and its equivalences $K = 1 + \chi|Q_d| = 1 + 1/(2N_c) = 1 + (B - L)/2 = Q + \Gamma$, are developed in Appendix C.

4.4 Chiral Activity Factors

Applying Eq. (18) to the LR isospin positions (8)–(10) gives:

$$\Gamma_{u,c,t} = 1 - \left(+\frac{1}{2}\right) = \mathbf{0.5000}, \quad (20)$$

$$\Gamma_{d,s} = 1 - \left(-\frac{1}{2}\right) = \mathbf{1.5000}, \quad (21)$$

$$\Gamma_b = 1 - \left(-\frac{1}{2} - \delta_b\right) = \frac{3}{2} + \delta_b. \quad (22)$$

The values of $\Gamma_{u,c,t}$ and $\Gamma_{d,s}$ are fixed entirely by the LR isospin-circle assignment and require no numerical fitting parameters. The b -quark factor contains the deformation δ_b , which is extracted from measured CKM 2–3 sector ratios (Section 5). The resulting activity factors are collected in Table 3.

Table 3: Weak isospin activity factors $\Gamma_q = 1 - T_3^R = 1 - T_3^L$. The first two rows are fixed by the LR isospin-circle assignment without any numerical fitting. The b -quark factor includes the deformation δ_b extracted from CKM 2–3 sector ratios (Section 5).

Quark(s)	T_3^R	$\Gamma_q = 1 - T_3^R$	Origin
u, c, t	$+\frac{1}{2}$	0.5000	LR isospin assignment (exact)
d, s	$-\frac{1}{2}$	1.5000	LR isospin assignment (exact)
b	$-\frac{1}{2} - \delta_b$	1.5665	CKM 2–3 sector extraction

5 Parameter Extraction from PDG Ratios

The construction uses three inputs extracted from *ratios* of adjacent-generation CKM elements. No individual element is used directly as an input. The generation-skip elements $|V_{ub}|$ and $|V_{td}|$ are not included in the extraction and serve as genuine predictions of the construction.

5.1 Zero-Parameter Test: Cabibbo Universality

The geometric degeneracy $\Gamma_d = \Gamma_s$, which follows from the shared T_3^R position of d_R and s_R (Table 1), is not an assumption adjusted to fit the data. It predicts the equality $|V_{us}|/|V_{ud}| = |V_{cd}|/|V_{cs}|$ with no free parameters. This relation can be tested directly against PDG 2024 values before any parameter is extracted:

$$\frac{|V_{us}|/|V_{ud}|}{|V_{cd}|/|V_{cs}|} = \frac{0.22500/0.97435}{0.22486/0.97349} = \frac{0.23092}{0.23098} = 0.99974. \quad (23)$$

The equality holds to 0.026%, confirming the shared T_3^R position of d_R and s_R without any tuning and validating the use of the geometric mean in the extraction below.

5.2 Extraction of Mixing Parameters

First-to-second generation mixing λ_{12} . Because $\Gamma_d = \Gamma_s$, the same mixing parameter describes the $(1 \leftrightarrow 2)$ transition in both the u -row and the c -row. The geometric mean of the two corresponding ratios gives:

$$\lambda_{12} = \sqrt{\frac{|V_{us}|}{|V_{ud}|} \cdot \frac{|V_{cd}|}{|V_{cs}|}} = \sqrt{\frac{0.22500}{0.97435} \cdot \frac{0.22486}{0.97349}} = \mathbf{0.23095}. \quad (24)$$

Second-to-third generation mixing λ_{23} . The analogous ratio in the 2–3 sector gives:

$$\lambda_{23} = \sqrt{\frac{|V_{cb}|}{|V_{cs}|} \cdot \frac{|V_{ts}|}{|V_{tb}|}} = \sqrt{\frac{0.04182}{0.97349} \cdot \frac{0.04110}{0.99912}} = \mathbf{0.04204}. \quad (25)$$

Bottom-quark amplitude factor Γ_b . The ratio of the two 2–3 sector mixing amplitudes gives Γ_b/Γ_s directly:

$$\frac{\Gamma_b}{\Gamma_s} = \frac{|V_{cb}|/|V_{cs}|}{|V_{ts}|/|V_{tb}|} = \frac{0.042959}{0.041136} = 1.0443. \quad (26)$$

Using the geometrically fixed value $\Gamma_s = 1.5000$:

$$\Gamma_b = 1.5000 \times 1.0443 = \mathbf{1.5665}, \quad \delta_b = \Gamma_b - \frac{3}{2} = \mathbf{0.0665}. \quad (27)$$

All PDG values used above are from Ref. [8].

Wolfenstein λ from λ_{12} . Within the LR isospin parametrisation, the Wolfenstein Cabibbo parameter λ is identified with the sine of the mixing angle whose tangent is λ_{12} :

$$\lambda = \frac{\lambda_{12}}{\sqrt{1 + \lambda_{12}^2}} = \frac{0.23095}{\sqrt{1.053338}} = \mathbf{0.22503}. \quad (28)$$

This agrees with the PDG 2024 global-fit value $\lambda = 0.22501$ to 0.009% [8], showing that λ is not an additional independent input but the geometric sine associated with the LR isospin transition parameter λ_{12} .

The three extracted quantities $\lambda_{12} = 0.23095$, $\lambda_{23} = 0.04204$, and $\Gamma_b = 1.5665$ are used throughout the subsequent CKM construction.

6 Anisotropic LR Isospin Metric and Geometric Action Scale

The characteristic scale of the LR isospin deformation is set by the b -quark amplitude factor extracted in Section 5.

6.1 Geometric Action Scale

The *unnormalised geometric action scale* is defined as:

$$s_{\text{unnorm}} \equiv \frac{\Gamma_b}{2} = \frac{3/2 + \delta_b}{2} = \mathbf{0.78325}. \quad (29)$$

After including the horizontal flavour normalisation associated with the Cabibbo rotation, the effective geometric action scale becomes:

$$s = \left(1 - \frac{\lambda^2}{2}\right) \frac{\Gamma_b}{2} = 0.97468 \times 0.78325 = \mathbf{0.76342}. \quad (30)$$

The factor $(1 - \lambda^2/2)$ is the standard second-order Cabibbo normalisation [5]. In the present formulation it connects the horizontal flavour-mixing contribution to the third-generation LR isospin deformation scale.

6.2 Anisotropic LR Isospin Metric and Constitutive Constraint

When $\delta_b \neq 0$, the circular LR isospin symmetry Eq. (7) is broken and replaced by an anisotropic LR isospin metric:

$$\frac{(T_3^L)^2}{a^2} + \frac{(T_3^R)^2}{b^2} = 1. \quad (31)$$

The deformation originates on the T_3^R axis: since $T_3^R(b_R) = -(\frac{1}{2} + \delta_b)$, the right-handed semi-axis expands to

$$b(\delta_b) = \frac{1}{2} + \delta_b = \mathbf{0.56650}. \quad (32)$$

The left-handed semi-axis is then fixed by the *linear constitutive constraint*

$$a + 2b = \frac{3}{2} = \Gamma_{d,s}, \quad (33)$$

which gives $a = \frac{3}{2} - 2b = \frac{1}{2} - 2\delta_b$:

$$a(\delta_b) = \frac{1}{2} - 2\delta_b = \mathbf{0.36700}, \quad b(\delta_b) = \frac{1}{2} + \delta_b = \mathbf{0.56650}. \quad (34)$$

The constraint (33) expresses the conservation of first-order gauge weight: the reduction of the T_3^L contribution (compressing a) is exactly balanced by the increase along the orthogonal T_3^R direction (expanding b), while the total is conserved at the value $\Gamma_{d,s} = \frac{3}{2}$ that characterises the unperturbed symmetric chiral sector. Note that a is not independently chosen: given b from the T_3^R deformation and the conservation constraint, a is uniquely determined.

6.3 Generation-Skip Projection Factors

In the symmetric limit $\delta_b = 0$, the LR isospin circle is undisturbed and the circle radius itself provides the common projection amplitude for both generation-skip directions:

$$\Phi_{13} = \Phi_{31} = |T_3| = \frac{1}{2}. \quad (35)$$

The two projections are equal in the symmetric case; no forward–backward asymmetry is generated.

The b -quark deformation introduces a directional asymmetry. The forward generation-skip amplitude Φ_{13} is identified with the normalised geometric action scale. The Wolfenstein apex $(\bar{\rho}, \bar{\eta})$ maps to a scaled image of the LR isospin circle via $\bar{\rho} + i\bar{\eta} = s T_3$, where s is the geometric action scale of Eq. (30) and $|T_3| = \frac{1}{2}$ on the circle. Taking the modulus gives $|\bar{\rho} + i\bar{\eta}| = s/2$, so the forward projection is:

$$\Phi_{13} = \frac{s}{2} = \mathbf{0.38171}. \quad (36)$$

The reverse projection ($t \rightarrow d$) acquires an additional orientation-dependent contribution from the deformed third-generation state. The effective LR isospin coordinate governing the reverse direction is:

$$T_3^{L,\text{eff}} = \left(1 - \frac{\lambda^2}{2}\right)^2 \frac{\Gamma_b(2 - \Gamma_b)}{4} = 0.94998 \times 0.16977 = \mathbf{0.16128}. \quad (37)$$

Here $\Gamma_b(2 - \Gamma_b) = (\frac{3}{2} + \delta_b)(\frac{1}{2} - \delta_b) = \frac{3}{4} - \delta_b - \delta_b^2$ is the bounded activity product of the deformed b -quark state: it reduces to $\frac{3}{4}$ in the symmetric limit $\delta_b \rightarrow 0$, and the factor $(1 - \lambda^2/2)^2$ carries the second-order Cabibbo normalisation for both the forward and backward projections. Substituting into

$$\Phi_{31}^2 = \Phi_{13}^2 + (1 - 2 T_3^{L,\text{eff}}) \quad (38)$$

gives the backward projection factor:

$$\Phi_{31} = \sqrt{\Phi_{13}^2 + (1 - 2 T_3^{L,\text{eff}})} = \sqrt{0.14570 + 0.67744} = \mathbf{0.90728}. \quad (39)$$

Equation (38) separates the symmetric LR isospin contribution Φ_{13}^2 (from the circle radius, Eq. 35) from the deformation-induced orientation asymmetry $(1 - 2T_3^{L,\text{eff}})$. The latter term distinguishes the forward and backward generation-skip projections and provides the geometric source of CP-sensitive structure in the CKM construction.

7 The CKM Matrix Construction

7.1 Flavour-Transition Suppression

A transition between non-adjacent generations can be represented as a sequential cascade through the intermediate generation: $(1 \rightarrow 3) = (1 \rightarrow 2) \otimes (2 \rightarrow 3)$. This sequential structure defines the *compositional suppression*:

$$S_{13} = S_{12} S_{23} = \lambda_{12} \lambda_{23}. \quad (40)$$

Substituting the values from Eqs. (24) and (25):

$$S_{13} = 0.23095 \times 0.04204 = \mathbf{0.0097091}. \quad (41)$$

The two-generation suppression is thus not an independent parameter but a geometric consequence of the two-step cascade. The full suppression table is:

$$S_{ij} = \begin{cases} 1 & |i - j| = 0, \\ \lambda_{12} & (i, j) = (1, 2) \text{ or } (2, 1), \\ \lambda_{23} & (i, j) = (2, 3) \text{ or } (3, 2), \\ \lambda_{12}\lambda_{23} & |i - j| = 2. \end{cases} \quad (42)$$

7.2 Matrix Generation Expression

Combining the chiral amplitude factors, the flavour-transition suppression, and the generation-skip projection factors, the magnitude of each CKM element is:

$$|V_{ij}| = \frac{\sqrt{\Gamma_i \Gamma_j} S_{ij} C_{ij}}{\mathcal{N}_i}, \quad (43)$$

where $\sqrt{\Gamma_i \Gamma_j}$ is the geometric mean of the chiral amplitude factors of the up-type quark i and down-type quark j , measuring the intrinsic coupling overlap in the (T_3^L, T_3^R) plane; S_{ij} accounts for the flavour-transition suppression between generations (Eq. 42); and the CP projection factors C_{ij} are:

$$C_{ij} = \begin{cases} \Phi_{13} = 0.38171 & (i, j) = (1, 3), \\ \Phi_{31} = 0.90728 & (i, j) = (3, 1), \\ 1 & |i - j| \leq 1. \end{cases} \quad (44)$$

The row normalisation factors $\mathcal{N}_i$ ($i = 1, 2, 3$) enforce unitarity for each row:

$$\mathcal{N}_i = \left(\sum_{j=1}^3 \Gamma_i \Gamma_j S_{ij}^2 C_{ij}^2 \right)^{1/2}, \quad (45)$$

ensuring $\sum_{j=1}^3 |V_{ij}|^2 = 1$ for $i = 1, 2, 3$.

7.3 CKM Matrix Evaluation

Substituting the chiral amplitude factors Γ_q , the flavour-transition suppression factors S_{ij} , and the generation-skip projection factors C_{ij} into Eq. (43) yields the pre-normalisation amplitudes:

$$V_{\text{unnorm}} = \begin{pmatrix} 0.86603 & 0.20001 & 0.003280 \\ 0.20001 & 0.86603 & 0.037206 \\ 0.007629 & 0.036408 & 0.885014 \end{pmatrix}. \quad (46)$$

The row normalisation factors from Eq. (45) are:

$$\begin{aligned} \mathcal{N}_1 &= \sqrt{0.86603^2 + 0.20001^2 + 0.003280^2} = 0.88883, \\ \mathcal{N}_2 &= \sqrt{0.20001^2 + 0.86603^2 + 0.037206^2} = 0.88960, \\ \mathcal{N}_3 &= \sqrt{0.007629^2 + 0.036408^2 + 0.885014^2} = 0.88580. \end{aligned}$$

Dividing each row by the corresponding $\mathcal{N}_i$ via Eq. (45) gives the fully normalised CKM matrix:

$$V_{\text{CKM}} = \begin{pmatrix} 0.97435 & 0.22503 & 0.003690 \\ 0.22483 & 0.97350 & 0.04182 \\ 0.00861 & 0.04110 & 0.99912 \end{pmatrix}. \quad (47)$$

8 Comparison with PDG 2024

Table 4 compares all nine constituted magnitudes against PDG 2024 experimental values and uncertainties [8]. The elements $|V_{ub}|$ and $|V_{td}|$ (marked $\dagger$) were not used in any parameter extraction step and are genuine predictions of the construction. All other elements entered only through adjacent-generation ratios, not as absolute inputs.

Table 4: Constituted CKM magnitudes compared with PDG 2024 [8]. Columns: PDG 2024 central value; experimental 1σ uncertainty; this-work value; absolute deviation; deviation in units of σ_{exp} . Elements marked $\dagger$ are genuine predictions not used in parameter extraction.

Element	PDG 2024	σ_{exp}	This work	$ \Delta $	$ \Delta /\sigma_{\text{exp}}$
$ V_{ud} $	0.97435	0.00016	0.97435	0.000004	0.03σ
$ V_{us} $	0.22500	0.00066	0.22503	0.000025	0.04σ
$ V_{ub} ^\dagger$	0.003732	0.000082	0.003690	0.000042	0.51σ
$ V_{cd} $	0.22486	0.00067	0.22483	0.000030	0.05σ
$ V_{cs} $	0.97349	0.00016	0.97350	0.000010	0.06σ
$ V_{cb} $	0.04182	0.00085	0.04182	0.000003	0.00σ
$ V_{td} ^\dagger$	0.00857	0.00020	0.00861	0.000042	0.21σ
$ V_{ts} $	0.04110	0.00083	0.04110	0.000002	0.00σ
$ V_{tb} $	0.99912	0.00013	0.99912	0.000002	0.02σ
<i>Maximum deviation across all nine elements:</i>					0.51σ

All nine magnitudes agree with PDG 2024 to within 0.51σ . The two genuine predictions are $|V_{ub}|$ (0.51σ) and $|V_{td}|$ (0.21σ). The seven elements used for parameter extraction are reproduced

to better than 0.06σ by construction. The diagonal elements $|V_{ud}|$, $|V_{cs}|$, and $|V_{tb}|$ were not used as direct inputs; they emerge from the row normalisation and agree with PDG 2024 to better than 0.06σ . For reference, the matrix of signed deviations (this work minus PDG 2024) is:

$$\delta V = \begin{pmatrix} -0.000004 & +0.000025 & -0.000042 \\ -0.000030 & +0.000010 & +0.000003 \\ +0.000042 & +0.000002 & -0.000002 \end{pmatrix}, \quad (48)$$

with all entries at the level of 10^{-5} , consistent with sub- 0.5σ agreement across the entire matrix.

9 Discussion

9.1 Physical Role of T_3^R

The key distinguishing feature of the present construction is the use of the right-handed weak-isospin coordinate T_3^R as an internal geometric anchor for right-handed quark states. In the Standard Model, right-handed quarks are singlets under $SU(2)_L$ and have $T_3^L = 0$; they possess no internal geometric anchor in the weak-isospin space. In left-right symmetric extensions, $T_3^R = \pm\frac{1}{2}$ provides this anchor. The amplitude factor $\Gamma_q = 1 - T_3^R$ then has a direct geometric reading, measuring the relative orientation of the right-handed state with respect to the co-aligned configuration ($T_3^R = +\frac{1}{2}$, up-type, $\Gamma < 1$) and the anti-aligned configuration ($T_3^R = -\frac{1}{2}$, down-type, $\Gamma > 1$). Up-type quarks, aligned with the positive T_3^R direction, couple weakly; down-type quarks, anti-aligned, couple strongly. This is the geometric origin of the leading-order scale $|V_{ij}|^2 \propto \Gamma_i \Gamma_j \sim \frac{1}{2} \times \frac{3}{2} = \frac{3}{4}$ that governs overall CKM mixing.

9.2 Geometric Origin of CP Violation

The undeformed LR isospin circle $|T_3| = \frac{1}{2}$ possesses full $U(1)$ rotational symmetry in the (T_3^L, T_3^R) plane. A construction based only on the radial invariant $|T_3|$ cannot distinguish opposite orientations and generates no forward-backward CP asymmetry.

CP-sensitive asymmetry enters when the LR isospin deformation $\delta_b > 0$ assigns physical quark states to *specific fixed points* on the circle: left-handed states at $(T_3^L, T_3^R) = (\pm\frac{1}{2}, 0)$ and right-handed states at $(0, \pm\frac{1}{2})$. This assignment breaks the $U(1)$ symmetry of the circle and produces $\Phi_{31} \neq \Phi_{13}$ via the decomposition:

$$\Phi_{31}^2 = \Phi_{13}^2 + (1 - 2T_3^{L,\text{eff}}), \quad (49)$$

where the first term is rotationally symmetric (circle-radius dependent) and the second, $(1 - 2T_3^{L,\text{eff}})$, is not: it singles out the T_3^L direction and distinguishes forward from backward generation-skip amplitudes. At the values of the present construction: $\Phi_{13} = 0.38171$, $\Phi_{31} = 0.90728$ (the forward $1 \rightarrow 3$ and backward $3 \rightarrow 1$ generation-skip projections defined in Eqs. (36)–(39); recall that in the symmetric limit $\delta_b \rightarrow 0$ both reduce to $\frac{1}{2}$, and $T_3^{L,\text{eff}} = 0.161$ (Eq. (37)) is the effective reverse-projection coordinate that breaks the forward-backward symmetry), giving the ratio $\Phi_{31}/\Phi_{13} = 2.376$, which reflects the forward-backward asymmetry of the CKM unitarity triangle.

The corresponding geometric estimates for the Wolfenstein CP parameters are obtained by identifying the apex coordinates with the internal geometric projections. With $\bar{\rho}_{\text{geom}} = T_3^{L,\text{eff}} = 0.161$, the companion coordinate $\bar{\eta}_{\text{geom}}$ follows from the constraint that the Wolfenstein apex lies on a circle of radius $s_{\text{unnorm}}/2 = \Gamma_b/4 = 0.39163$ in the barred-parameter plane:

$$\bar{\eta}_{\text{geom}} = \sqrt{\left(\frac{\Gamma_b}{4}\right)^2 - \bar{\rho}_{\text{geom}}^2} = \sqrt{(0.39163)^2 - (0.16128)^2} = \sqrt{0.12736} = 0.357. \quad (50)$$

Comparing with PDG 2024 [8]: $\bar{\rho} = 0.159$ (1.3% accuracy) and $\bar{\eta} = 0.348$ (2.6% accuracy). These first-order geometric estimates confirm that the LR isospin deformation accounts for the qualitative CP structure; exact agreement would require a higher-order treatment of the deformation dynamics.

CP activity deviation. The deformation parameter δ_b admits a direct interpretation in terms of the activity operator. In the symmetric LR configuration, both the d and s quarks carry $T_3^R = -\frac{1}{2}$, giving the common activity $\Gamma_{d,s} = \Gamma_0 \equiv \frac{3}{2}$ for the down-type sector. The b quark departs from this reference:

$$\Gamma_b = \Gamma_0 + \Delta\Gamma_{\text{CP}}, \quad \Delta\Gamma_{\text{CP}} \equiv \Gamma_b - \frac{3}{2} = \delta_b, \quad (51)$$

so $\Delta\Gamma_{\text{CP}} = 0.0665$ at the PDG-extracted value. The quantity $\Gamma_0 = \frac{3}{2}$ is the CP-symmetric activity of the down-type sector; $\Delta\Gamma_{\text{CP}} > 0$ measures the departure that breaks the left-right degeneracy of the d, s, b triplet. Since CP violation in the CKM matrix requires three generations, the appearance of $\Delta\Gamma_{\text{CP}}$ exclusively for the third-generation down-type quark is consistent with the three-generation origin of the CP-violating phase. A first-principles derivation of $\Delta\Gamma_{\text{CP}}$ from the LR weak-isospin geometry — presently given by the constitutive saturation law (Appendix A) — remains an objective for future investigation.

9.3 Parameter Reduction and Predictive Scope

The present construction generates all nine CKM magnitudes from two extracted quantities, λ_{12} and λ_{23} , obtained from PDG 2024 adjacent-generation ratios [8]. The third quantity, Γ_b (equivalently δ_b), is not an independent external input: it is determined by the constitutive saturation relation derived in Appendix A, Level 2. These quantities are not adjusted to fit the two generation-skip predictions $|V_{ub}|$ and $|V_{td}|$, which therefore serve as genuine predictions of the construction.

The relation (28), $\lambda = \lambda_{12}/\sqrt{1 + \lambda_{12}^2}$, shows that the Wolfenstein Cabibbo parameter is not a third independent input: it is the geometric sine associated with the LR isospin transition parameter λ_{12} , in agreement with PDG 2024 to 0.009% [8]. The Wolfenstein apex $(\bar{\rho}, \bar{\eta})$ is moreover linked to the LR isospin geometry via $\bar{\rho} + i\bar{\eta} = s T_3$ (see Section 6), providing a direct geometric interpretation of the CP structure.

A further reduction of the two extracted inputs requires a dynamical derivation of λ_{12} and λ_{23} from the underlying LR isospin deformation dynamics. Possible routes are discussed in Appendix A.

10 Conclusion

We have constructed the CKM matrix magnitudes using the geometry of left-right symmetric weak-isospin coordinates. The key feature is that quark flavour-mixing magnitudes emerge from geometric relations in the (T_3^L, T_3^R) internal quantum plane, rather than being specified solely as independent phenomenological parameters.

The main elements of the derivation are: (i) the LR isospin circle $(T_3^L)^2 + (T_3^R)^2 = \frac{1}{4}$, where T_3^R is the right-handed $\text{SU}(2)_R$ weak-isospin projection that provides geometric anchor points for right-handed quarks; (ii) the weak-isospin activity operator $\hat{\Gamma} = \mathbb{I} - \hat{T}_3^L$, which yields exact symmetric-sector amplitude factors $\Gamma_{u,c,t} = \frac{1}{2}$ and $\Gamma_{d,s} = \frac{3}{2}$ without additional parameters; (iii) a single constitutive deformation parameter $\delta_b = 0.0665$ governing the bottom-quark departure from the LR-symmetric configuration; (iv) the linear constitutive constraint $a + 2b = \frac{3}{2}$, expressing first-order conservation of internal gauge weight under LR isospin deformation; and (v) compositional suppression $S_{13} = S_{12}S_{23}$, showing that the two-generation skip amplitude is not an independent transition parameter.

Two inputs (λ_{12} , λ_{23}) extracted from PDG 2024 adjacent-generation ratios [8], combined with the constitutive saturation law derived in the Appendix that determines Γ_b geometrically, suffice to reproduce all nine CKM magnitudes within 0.51σ . The two non-input predictions $|V_{ub}|$ and $|V_{td}|$ agree with experiment at 0.51σ and 0.21σ , respectively.

Two supplementary derivations (Appendices A and B) extend the construction toward a fully predictive framework. Appendix A derives δ_b analytically from a constitutive saturation law, $\delta_b(1 - \delta_b) = \frac{7}{6}\lambda_{12}^2$, obtained from the LR charge formula for the down-type quark; the result $\delta_b = 0.06667$ agrees with the PDG-extracted value to 0.01%, confirming internal consistency and reducing the independent external inputs from three to two. Appendix B establishes that the Wolfenstein sine parameter $\lambda = 1/(\pi\sqrt{2}) \approx 0.22508$ coincides with the azimuthal projection of the LR isospin unit circle onto its CP-neutral axis, reproducing the PDG value to 0.03% with no CKM input. A numerical check confirms that substituting λ_{geom} for λ_{kin} shifts every CKM element by less than 0.001%, leaving the 0.51σ agreement unchanged. If the geometric origin of λ is substantiated, the number of independent external inputs reduces further from two to one (λ_{23} alone), bringing a zero-input fully predictive construction within reach.

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A Parameter Accounting and Reduction Roadmap

Table 5 summarises the parameters entering the CKM construction by origin and status.

Level 1 (direct extraction): 3 inputs. The CKM magnitudes can be obtained from λ_{12} , λ_{23} , and Γ_b , all extracted from PDG 2024 adjacent-generation ratios (Section 5). At this level $\Gamma_b = 1.5665$ is treated as an independently extracted input; the table above reflects the Level 2 status in which $\Gamma_b = 1.5667$ is derived. These quantities are not adjusted to reproduce the predicted elements $|V_{ub}|$ and $|V_{td}|$, which serve as independent tests. This level serves as a self-consistency check for Level 2.

Level 2 (constitutive saturation law): 2 inputs. The deformation parameter δ_b is determined analytically from a constitutive saturation condition relating the right-handed isospin deformation of the b quark to the electric charge structure of the down-type quark.

Table 5: Parameter inventory by tier. Tier 1: exact from LR isospin geometry. Tier 2: extracted from PDG 2024 adjacent-generation ratios [8]. Tier 2': derived via the Level 2 saturation law (this Appendix); $\Gamma_b = 1.5667$ (saturation) agrees with the Level 1 extracted value $\Gamma_b = 1.5665$ (Section 5) to 0.01%. Tier 3: derived identities, not independent inputs.

Parameter	Value	Formula	Tier
$\Gamma_{u,c,t}$	0.5000	$1 - T_3^R$, LR circle	1 — exact
$\Gamma_{d,s}$	1.5000	$1 - T_3^R$, LR circle	1 — exact
λ_{12}	0.23095	$\sqrt{(V_{us} / V_{ud})(V_{cd} / V_{cs})}$	2 — extracted
λ_{23}	0.04204	$\sqrt{(V_{cb} / V_{cs})(V_{ts} / V_{tb})}$	2 — extracted
Γ_b	1.5667	$\frac{3}{2} + \delta_b$, δ_b from Eq. (54)	2' — derived (this Appendix)
λ	0.22503	$\lambda_{12}/\sqrt{1 + \lambda_{12}^2}$	3 — derived
s, Φ_{13}, Φ_{31}	see text	from λ and Γ_b	3 — derived

Physical motivation. The deformation δ_b displaces $T_3^R(b_R)$ from the symmetric doublet value $-\frac{1}{2}$ to $-(\frac{1}{2} + \delta_b)$. The saturation product $\delta_b(1 - \delta_b)$ measures the overlap between the deformed and symmetric isospin configurations: it vanishes at $\delta_b = 0$ (no deformation) and at $\delta_b = 1$ (complete saturation), with maximum at $\delta_b = \frac{1}{2}$.

By the Born rule of quantum mechanics, the transition probability corresponding to the leading Cabibbo mixing amplitude λ_{12} is

$$P_{12} = |\lambda_{12}|^2 = \lambda_{12}^2. \quad (52)$$

The b -quark deformation δ_b is a structural consequence of this dominant first-to-second generation mixing and is therefore driven by P_{12} at leading order. The higher-generation transition probabilities satisfy $\lambda_{23}^2/\lambda_{12}^2 \approx 0.033$ and $\lambda_{13}^2/\lambda_{12}^2 \approx 0.002$; their contributions are suppressed by factors of 30 and 500 respectively and do not enter at leading order.

The Background Stabilization Factor $K = 7/6$ (Appendix C) converts the Born probability P_{12} into the geometric deformation of the b -quark right-handed isospin coordinate via $\delta_b(1 - \delta_b) = K P_{12}$. Because $K > 1$, the electroweak background amplifies the structural deformation beyond the naive Born probability; the excess $K - 1 = \frac{1}{6} = (B - L)/2$ is traceable to the $U(1)_{B-L}$ sector.

Saturation law and its solution. The constitutive saturation law is

$$\delta_b(1 - \delta_b) = K\lambda_{12}^2 = \frac{7}{6}\lambda_{12}^2, \quad (53)$$

a quadratic in δ_b whose physical (smaller) root is

$$\delta_b = \frac{1 - \sqrt{1 - \frac{14}{3}\lambda_{12}^2}}{2}. \quad (54)$$

Numerical verification. Inserting $\lambda_{12} = 0.23095$:

$$\begin{aligned} \frac{7}{6}\lambda_{12}^2 &= \frac{7}{6} \times (0.23095)^2 = 1.16\bar{6} \times 0.053338 = 0.062228, \\ 1 - 4 \times 0.062228 &= 0.75109, \quad \sqrt{0.75109} = 0.86665, \\ \delta_b &= \frac{1}{2}(1 - 0.86665) = \mathbf{0.06667}, \\ \Gamma_b &= \frac{3}{2} + 0.06667 = \mathbf{1.56667}. \end{aligned} \quad (55)$$

This agrees with the PDG-extracted value $\Gamma_b = 1.5665$ (Section 5) to **0.01%** — an improvement of a factor of eight over the first-order linear estimate at Level 1 (0.08%). Since δ_b is now determined by λ_{12} alone through Eqs. (53)–(54), the number of independent extracted inputs reduces from three to two (λ_{12} and λ_{23}).

Level 3 (quark-mass hierarchy): 1 input. The ratio $\lambda_{23}/\lambda_{12} = 0.04204/0.23095 = 0.182$ shows numerical proximity to the hierarchy between successive down-type quark mass gaps. A derivation of λ_{23} from the underlying quark mass spectrum would reduce the number of external inputs to one.

Long-term objective: 0 independent inputs. The remaining Cabibbo-sector parameter λ_{12} (the tangent of the Cabibbo angle) may ultimately be connected to the dynamics of LR symmetry breaking, providing a route towards a fully predictive geometric construction of quark flavour mixing. A supplementary geometric observation pointing in this direction is presented in Appendix B.

B Geometric Observation on Wolfenstein Sine Parameter λ

This appendix shows a geometric observation that connects the Wolfenstein sine parameter λ directly to the LR weak-isospin circle, providing an independent route to the Cabibbo scale without fitting to CKM data.

Kinematic identity

The internal mixing parameter used throughout the main text is the *mixing tangent*

$$\lambda_{12} \equiv \tan \theta_{12} = \sqrt{\left(\frac{|V_{us}|}{|V_{ud}|}\right)\left(\frac{|V_{cd}|}{|V_{cs}|}\right)} = 0.23095, \quad (56)$$

extracted from ratios of adjacent CKM elements (Section 5). The Wolfenstein expansion parameter $\lambda \equiv \sin \theta_{12}$ follows from the exact kinematic identity

$$\lambda = \frac{\lambda_{12}}{\sqrt{1 + \lambda_{12}^2}} = \frac{0.23095}{\sqrt{1 + (0.23095)^2}} = 0.22503. \quad (57)$$

This identity converts the geometric mean of CKM column ratios (a tangent) to the Wolfenstein expansion parameter (a sine). The current PDG value is $\lambda_{\text{PDG}} = 0.22501$; the kinematic identity reproduces it to 0.009%.

Geometric prediction from the LR isospin circle

The LR isospin circle $(T_3^L)^2 + (T_3^R)^2 = \frac{1}{4}$ has radius $\frac{1}{2}$. Rescaling to unit radius defines the normalized coordinates $(\tilde{\rho}, \tilde{\eta}) \equiv (2T_3^L, 2T_3^R)$, which satisfy $\tilde{\rho}^2 + \tilde{\eta}^2 = 1$. At the $\text{SU}(2)_L\text{-SU}(2)_R$ -symmetric point $(\tilde{\rho}, \tilde{\eta}) = (1/\sqrt{2}, 1/\sqrt{2})$, where left and right projections are equal, $\tilde{\rho} = 1/\sqrt{2}$. Dividing this symmetric projection by the half-space aperture π (the azimuthal range spanned by the $T_3^L > 0$ sector on the unit circle) gives

$$\lambda_{\text{geom}} = \frac{\tilde{\rho}}{\pi} = \frac{1}{\pi\sqrt{2}} \approx 0.22508. \quad (58)$$

This expression matches the kinematic identity Eq. (57) to **0.023%** and the PDG value to **0.03%**, without any CKM input. The agreement supports the interpretation of λ as the fundamental azimuthal projection scale of the LR complex coordinate $T_3 = T_3^L + iT_3^R$.

We note that the formula (58) is presented here as a geometric *observation*: the derivation of $\tilde{\rho} = 1/\sqrt{2}$ from the LR symmetry-breaking dynamics is left to future work.

Consistency check with the Wolfenstein parametrization

Using $\lambda_{\text{geom}} = 1/(\pi\sqrt{2})$ and the PDG Wolfenstein parameter $A = 0.826$ [8], the second-generation off-diagonal element is

$$|V_{cb}|_W = A\lambda_{\text{geom}}^2 = 0.826 \times \left(\frac{1}{\pi\sqrt{2}}\right)^2 = 0.04185, \quad (59)$$

which agrees with the PDG central value 0.04182 ± 0.00085 to 0.03σ .

For the third-generation transition, the Wolfenstein formula requires the full apex factor $R_b \equiv \sqrt{\bar{\rho}^2 + \bar{\eta}^2} \approx 0.383$ (where $\bar{\rho}$ and $\bar{\eta}$ are the standard Wolfenstein CP-violation parameters):

$$|V_{ub}|_W = A\lambda_{\text{geom}}^3 R_b = 0.826 \times (0.22508)^3 \times 0.383 = 0.00360. \quad (60)$$

The PDG value is $|V_{ub}| = 0.003732 \pm 0.000082$, giving a pull of 1.6σ from Eq. (60). This is larger than the 0.5σ agreement achieved by the master formula Eq. (43) in Section 8. The discrepancy reflects the fact that $|V_{ub}|$ is a generation-skip element that receives its principal contribution from the chiral amplitude product $\sqrt{\Gamma_t \Gamma_b}$ and the generation-skip projections Φ_{13}, Φ_{31} , which are constructed geometrically in the main text and are absent in the standard Wolfenstein expansion. The Wolfenstein formula should therefore be regarded as a rough cross-check for $|V_{ub}|$, not as a competing derivation.

C Derivation of the Background Stabilization Factor $K = 7/6$

This appendix derives the Background Stabilization Factor $K = 7/6$ from the charge geometry of the T_3^R axis, and collects its equivalent expressions.

Charge structure on the T_3^R axis. In the LR charge formula (5), the electric charge of the down-type quark is

$$Q_d = T_3^L + T_3^R + \frac{B-L}{2} = -\frac{1}{2} + \left(-\frac{1}{2}\right) + \frac{1}{6} = -\frac{1}{3}, \quad |Q_d| = \frac{1}{3}. \quad (61)$$

(Here $B-L = \frac{1}{3}$ for each quark.) The T_3^R doublet axis spans the symmetric range $[-\frac{1}{2}, +\frac{1}{2}]$.

Overlap factor. The coupling between the isospin deformation and the electric charge is mediated by the characteristic overlap factor between the boundary and midpoint of the doublet:

$$\chi = \frac{|T_3^R|_{\text{boundary}}}{|T_3^R|_{\text{boundary}} + |T_3^R|_{\text{midpoint}}} = \frac{\frac{1}{2}}{\frac{1}{2} + \frac{1}{2}} = \frac{1}{2}. \quad (62)$$

This factor selects the fraction of the doublet range that is charge-active: the boundary value $|T_3^R| = \frac{1}{2}$ carries the full charge coupling, while the midpoint $|T_3^R| = 0$ is charge-neutral.

Derivation of K . The Background Stabilization Factor combines the unit baseline (the LR isospin circle contribution) with the charge-weighted coupling:

$$K = 1 + \chi |Q_d| = 1 + \frac{1}{2} \cdot \frac{1}{3} = \frac{7}{6}. \quad (63)$$

This is not a free parameter: both $\chi = \frac{1}{2}$ and $|Q_d| = \frac{1}{3}$ are fixed by the LR isospin geometry and the quark charge assignment.

Equivalent expressions. Since the quark charge magnitude $|Q_d| = \frac{1}{3} = \frac{1}{N_c}$ (where $N_c = 3$ is the number of colour charges, as follows from $B = \frac{1}{N_c}$ and the LR charge formula), Eq. (63) can be written compactly in terms of the colour-charge content:

$$K = 1 + \frac{\chi}{N_c} = 1 + \frac{1}{2N_c} = \frac{7}{6}. \quad (64)$$

Since $(B - L)/2 = \frac{1}{6}$ for quarks (with $B = \frac{1}{3}$, $L = 0$), the increment $\chi|Q_d| = \frac{1}{6}$ is precisely the baryon-number hypercharge contribution in the LR charge formula, giving

$$K = 1 + \frac{B - L}{2} = \frac{7}{6}. \quad (65)$$

This links K directly to the $U(1)_{B-L}$ sector of the LR gauge group.

GNN universality. The four expressions $K = 1 + \chi|Q_d| = 1 + \frac{1}{2N_c} = 1 + \frac{B-L}{2} = Q + \Gamma$ are all equivalent. Their agreement with the algebraic identity Eq. (14) and the numerical universality $K = 7/6$ in Table 2 confirms that the Background Stabilization Factor is uniquely determined by the GNN charge structure of the quark sector and admits no free variation.